
ZOÉ BERENGER

PhD in Applied Mathematics and Image Processing

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	2021–2025	PhD: Deep learning and SAR tomography for forest structure monitoring. TÉLÉCOM PARIS, IP Paris, Palaiseau (91). Under the supervision of Florence Tupin, Laurent Ferro-Famil, and Loïc Denis. Teledetection and environmental modelling of scene regularity via a network learning from radar measurements in the ESA BIOMASS satellite mission context. 2-month research exchange at DLR, Germany, supervised by Andreas Reigber.
	2019–2020	MSc in Computing. NATIONAL UNIVERSITY OF SINGAPORE. Double Diploma as a Graduate student in the School of Computing. Artificial Intelligence (ML, DL), Spatial Big Data and Analytics courses and master thesis.
Skills PROGRAMMING LANGUAGES Python (PyTorch, NumPy, Scikit-learn, TensorFlow), C++, SQL (PostgreSQL), Java GEOSPATIAL & SCIENTIFIC TOOLS ArcGIS, OpenGL, slurm, Linux, LaTeX DATA ENGINEERING & INFRASTRUCTURE Docker, Git WEB & MOBILE DEVELOPMENT Android, JavaScript, HTML, CSS LANGUAGES French: native. English: C1 level. 2013–2014: Academic-year cultural exchange at RJR HIGH SCHOOL, Winston-Salem, NC, USA. TOEFL IBT 2018 score: 109/120. German: B2 level. Mars-Avril 2012: 2 months student exchange in SCHWÄBISCH HALL, Germany (program BRIGITTE-SAUZAY).	2017–2020	Engineer's degree. TÉLÉCOM PARIS, Paris 13. Courses in computer vision, machine learning, statistics, optimization, data science, computer graphics, human computer interaction and data visualization.
	2015–2017	Scientific preparatory classes. LYCÉE DU PARC, Lyon. Intensive preparation to national competitive exams in Physics and Mathematics.
	Working Experience	
	2021–2024	Teaching Assistant. TÉLÉCOM PARIS, IP Paris, Palaiseau (91). Supervision of practical work and projects in optical and radar image processing.
	Jan-Sept 2021	Product Manager. MANA COMMUNITY, Paris. Mission in an NGO that identifies incidents committed by companies on forests from Twitter and local press with machine-learning based models, contributing to the monitoring of over 400 companies (Pytorch, Pytest, JavaScript, DBEaver).
	July-Dec 2020	Software Engineer. QARNOT COMPUTING, Montrouge (92). End-of-year internship in a firm providing an eco-friendly HPC infrastructure. - Numerical weather prediction and deep learning software integration (WRF, Docker). - Natural language processing proof of concept for speech recognition (Pytorch).
	2017–2021	Mathematics and physics tutor. Middle and high school students.
	Research Projects	
	2019–2020	Master thesis: Generating art through human motion. Art generated through video style transfer using deep learning on optical flow videos derived from human movements (Processing, Keras, OpenCV).
	April 2020	Influencing factors of Chinese wildfires based on Big Data. Prediction of fire occurrences with logistic regressions (ArcGIS), identifying key environmental drivers for large-scale fire risk assessment. Group of 6.
	March 2020	Hierarchical learning for predicting cancerous cells proportions. Prediction with a novel hierarchical approach (Python, Keras). Work in pair.
	April 2019	Machine learning challenge for melanoma diagnosis. Features extraction on and classification of 1000 skin lesion. Kaggle challenge.
	June 2018	Brain's white fibres 3D modelling and visualization tool. Contributing to neural structure understanding (C++, OpenGL). Group of 3.
	Additional Experiences	
	2018–2019	Student associations. TÉLÉCOM PARIS. 2018–2019: HR and communications manager (Junior Enterprise), treasurer (Sustainable development), opera and ballet manager (Art and culture).
	2014–2015	BAFA, French Youth Leadership Certificate. Lifeguard specialisation.
